

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (4 points)

(a) Describe the column space of A , $C(A)$, where $A = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 0 \end{bmatrix}$.

$$\begin{aligned} AX &= \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + 2x_2 \\ x_1 + 2x_2 \\ 0 \end{bmatrix} \\ &= (x_1 + 2x_2) \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \\ &= c \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \\ \Rightarrow C(A) &= \left\{ c \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \mid c \text{ is any number} \right\} \end{aligned}$$

(b) Let A be an arbitrarily given matrix. True or false:

(i) The column space of $3A$ equals the column space of A .

True

(ii) The vectors b that are not in the column space $C(A)$ form a subspace.

False

Problem 2. (6 points) Let $A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ -2 & -6 & 0 & 0 \end{bmatrix}$.

(a) Reduce A to the echelon form U by elimination.

$$A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ -2 & -6 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 2 & 4 \\ -2 & -6 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

(b) Which are the free variables and which are the pivot variables?

Pivot variables: x_1, x_3

Free variables: x_2, x_4

(c) Find the null space of A , $N(A)$.

We need solve $UX = 0$ to find all special solutions

$$\begin{cases} x_1 + 3x_2 + x_3 + 2x_4 = 0 \\ 2x_3 + 4x_4 = 0 \end{cases}$$

Let $x_2 = 1, x_4 = 0$, we get

$$\begin{cases} x_1 + x_3 = -3 \\ 2x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = -3 \\ x_3 = 0 \end{cases} \Rightarrow s_1 = \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

Let $x_2 = 0, x_4 = 1$, we get

$$\begin{cases} x_1 + x_3 = -2 \\ 2x_3 = -4 \end{cases} \Rightarrow \begin{cases} x_1 = 0 \\ x_3 = -2 \end{cases} \Rightarrow s_2 = \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

Thus

$$N(A) = \left\{ c_1 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix} \mid c_1, c_2 \text{ any numbers} \right\}$$

$$= \left\{ \begin{bmatrix} -3c_1 \\ c_1 \\ -2c_2 \\ c_2 \end{bmatrix} \mid c_1, c_2 \text{ any numbers} \right\}$$